

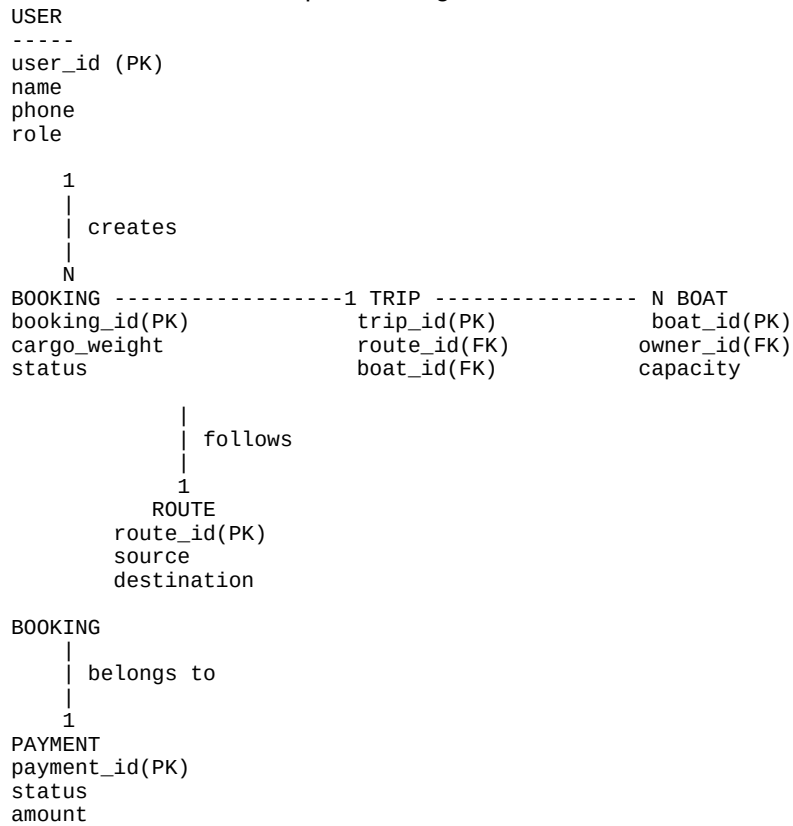
NoboGhat

System Design Document (Part 3)

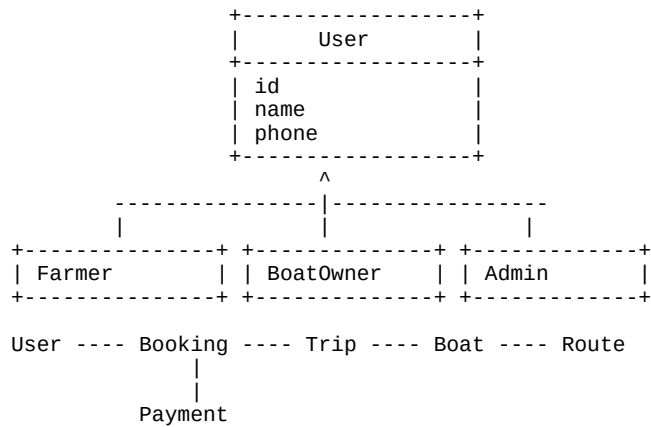
Database Design • OOP Class Diagram • Use Case Diagram • Activity Diagram • System Workflow

1. Database Design (Conceptual ER Model)

Core entities and their relationships are designed around real-world inland cargo operations.



2. OOP Class Diagram (Java Design)



Classes

User

Farmer extends User

BoatOwner extends User

Admin extends User

Boat

Route

Trip

Booking

Payment

Notification (future)

3. Use Case Diagram (Textual)

```
+-----+  
|      NoboGhat      |  
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```

Farmer

```
|-- Register  
|-- Login  
|-- Search Route  
|-- Book Cargo  
|-- View Booking
```

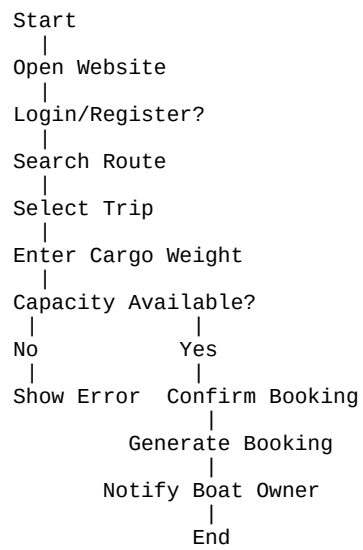
Boat Owner

```
|-- Register Boat  
|-- Create Trip  
|-- Update Capacity  
|-- View Bookings
```

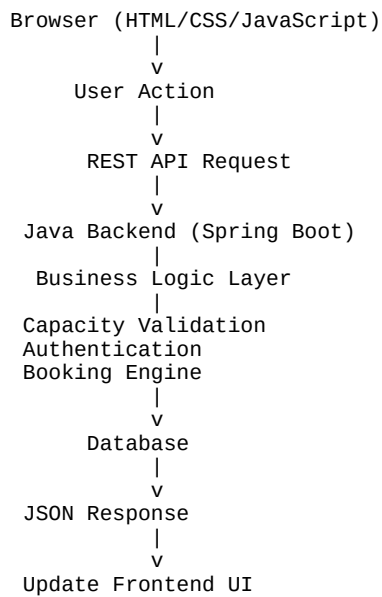
Administrator

```
|-- Verify Users  
|-- Manage Routes  
|-- Monitor System
```

4. Activity Diagram



5. Overall System Workflow



Design Notes

The architecture is intentionally layered. The frontend is responsible only for presentation, the backend contains business rules implemented using Java OOP classes, and the database stores persistent data. This separation makes the system easier to maintain, test, and expand into future features such as GPS tracking, digital payments, flood alerts, and logistics analytics.